

HITHA MAGADI VIJAYANAND

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EDUCATION

- Texas A&M University**
Master of Computer Science - GPA : 3.77/4.00
- Aug 2024 – May 2026
- JSS Science And Technology University**
Bachelor of Engineering Computer Science - GPA : 9.52/10.00
- Aug 2017 – Jul 2021

TECHNICAL SKILLS

Language & Frameworks : C, C++, Ruby, **Python, Java**, JavaScript, HTML, CSS, React, Spring Boot, Node, TypeScript, Node.js
Tools & Platforms : Git, Github, **AWS, Docker, Kubernetes**, Linux, Unix, OpenSearch, Terraform, Jenkins, Jira, Splunk
Database & Others : MongoDB, **NoSQL, SQL, MySQL**, PostgreSQL, **DynamoDB, Lambda, S3**, Machine Learning, REST API, Data Structures, Algorithms, NLP, Agile, CI/CD , SDLC, Networking, DevOps, Service Deployments, Distributed Systems, Operating System, Infrastructure As Code (Terraform) ,Containerization, RabbitMQ, AmazonMQ, Cloud Infrastructure, Grafana, Prometheus, SRE, HPC Environments, Kafka, Full-Stack, Data Modeling, GPU, Spark, Scala

PROFESSIONAL EXPERIENCE

- JP Morgan Chase**
Software Development Engineer 2
- Jan 2024 – Aug 2024
- Migrated microservices from a private to public cloud **AWS ECS**, **improving scalability** and **reducing deployment time** by 70%
 - Redesigned the user notification workflow using an **event-driven architecture** powered by **RabbitMQ**, enabling the platform to handle and process over **15 million events** per month without performance degradation
 - Developed a **real-time synonym training** system that fine-tuned NLP search models with evolving financial terminology, improving search accuracy by 30%
 - Integrated **AI-assisted development** workflows using GitHub Copilot to improve developer productivity
- JP Morgan Chase**
Software Development Engineer 1
- Aug 2021 – Dec 2023
- Led the end-to-end modernization of the platform by upgrading **Spring Boot** from **2.0 to 3.0** and **Java 8** to **Java 17**
 - Migrated a legacy MarkLogic system to a cloud-native architecture using AWS S3 and DynamoDB, improving scalability, resolving versioning issues and **reducing infrastructure costs by 80%**
 - Designed **ETL data pipelines** to ingest, transform, and **index 1M+ records** into **AWS OpenSearch**
 - Created **interactive dashboards** for 5+ teams, streamlining data access and enhancing operational transparency
- JP Morgan Chase**
Backend Development Intern
- Feb 2021 – Jul 2021
- Automated business report generation via Outlook and Control-M, ensuring timely delivery and reducing manual effort
 - Achieved 100% method coverage with **JUnit** tests and validated performance under load using **JMeter**
 - Developed scalable **Backend APIs** for creating, updating, and managing **OpenSearch indices**, streamlining data indexing workflows and enabling efficient configuration management and system administration

PROJECTS

- Hybrid Self-RAG - Hallucination Reduction In LLMs | NLP, RAG, OpenAI** 
- Jan 2025 – May 2025
- Built a **Hybrid Self-RAG** system using **Wikipedia** and **ArXiv APIs** improving factual grounding and accuracy by 9%
 - Constructed a robust **FactScore-Based** Validation framework utilizing **Sentence-Transformer** embeddings and **cosine similarity** metrics to enhance the factual reliability of LLM outputs, increasing the average FactScore from 0.65 to 0.87
- Generator Matrix Optimization | Evolutionary Algo, Mathematics** 
- Jan 2025 – May 2025
- Implemented a hybrid optimization pipeline combining **linear programming** with **genetic and simulated annealing algorithms** to minimize m-height in generator matrices for linear error-correcting codes
 - Enhanced encoding efficiency and fault tolerance for high-reliability communication and storage systems
- Multimodal Network - Advanced Calorie Prediction | CNN, RNN, LSTM** 
- Aug 2024 – Dec 2024
- Designed a multimodal deep learning architecture that integrates **Bi-LSTM** and **CNN** components to fuse CGM time-series data, neural signal features, and meal images, enabling more accurate and context-aware calorie prediction
 - Achieved **0.33 RMSRE**, outperforming baselines and validating multimodal integration in health ML models

CERTIFICATIONS AND ACHIEVEMENTS

- **Won JP Morgan's national-level hackathon** by developing a scalable solution to **transfer large media files** to **AWS S3** in Aug 2022
- **AWS Cloud Practitioner** certification issued by Amazon Web Services in Jul 2022 - Transferable to Azure